MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Core Course (DSC) - MG4DSCGEO201 - INVERTEBRATE PALEONTOLOGY
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Q1. Pseudofossil vs Synthetic fossil

- Pseudofossil:
    - Inorganic structure resembling fossils (0.5)
    - Formed by natural geological processes (0.5)
- Synthetic fossil:
    - Artificial/man-made replica (0.5)
    - Used for study/display (0.5)

Q2. Byssus in bivalves

- Definition: thread-like structure (0.5)
- Composition: proteinaceous secretion (0.5)
- Function: attachment to substrate (1)

Q3. Genital vs Ocular plates

- Genital plates:
    - Associated with reproductive openings (1)
- Ocular plates:
    - Associated with eye structures/light sensing (1)

Q4. Cast

- Definition: positive replica of organism (1)
- Formed by filling of mold with sediment/mineral (1)

Q5. Phylogeny

- Evolutionary history of organisms (1)
- Shows relationships among taxa (1)

Q6. Flora vs Fauna

- Flora: plant life (1)
- Fauna: animal life (1)

Q7. Vertebrate palaeontology

- Study of fossil vertebrates (1)
- Includes evolution, classification, morphology (1)

Q8. Calcareous microfossils (any two)

- Each correct example (Foraminifera / Coccoliths / Ostracods): 1 mark × 2

PART B (8 × 3 = 24 marks)

Q9. Contribution of Indian fossil flora

- Gondwana flora (Glossopteris, Gangamopteris) – 2
- Stratigraphic correlation – 1
- Paleoclimate reconstruction – 2
- Evolution of plant groups – 1
- Economic significance (coal deposits) – 1
- Critical evaluation (limitations/significance) – 1

Q10. Shell structure of brachiopods

- Composition (calcite/chitinophosphate) – 1
- Shell layers:
    - Primary – 1
    - Secondary – 1
    - Tertiary – 1
- Internal structures (brachidium, hinge) – 1
- Ornamentation – 1
- Functional significance – 1
- Diagram – 1

Q11. Fossilization of unaltered hard parts

- Definition – 1
- Conditions:
    - Rapid burial – 1
    - Low chemical alteration – 1
- Preservation examples – 2

- Process explanation – 2
- Clarity/organization – 1

Q12. Cephalopod ornamentation

- Types (ribs, spines, nodes) – 2
- Description of patterns – 1
- Hydrodynamic function – 2
- Protection role – 1
- Evolutionary/adaptive significance – 1
- Integration of concept – 1

Q13. Fossils as paleoclimate indicators

- Concept explanation – 2
- Floral indicators – 2
- Faunal indicators – 2
- Sedimentary association – 1
- Examples (corals, coal, evaporites) – 1

PART C (12 × 1 = 12 marks)

Q14. Gastropod coiling patterns

- Types:
    - Planispiral – 1
    - Conispiral – 1
    - Others (helicoidal, trochospiral) – 1
- Coiling mechanism – 2
- Functional significance:
    - Stability – 1
    - Protection – 1
- Evolutionary trends – 2
- Ecological adaptations – 2
- Diagram/clarity – 1

Q15. Microfossils & coal/petroleum

- Definition of microfossils – 1
- Types (forams, pollen, spores) – 2
- Stratigraphic utility – 2
- Petroleum exploration:
    - Age determination – 1
    - Environmental interpretation – 1
- Coal formation indicators – 2
- Analytical integration – 2
- Clarity – 1